

Supplementary Material for:
Repetition, Not Length: Isolating the Counting
Failure in Neural Text-to-Speech

Anonymous ICASSP submission

This document accompanies the 4-page ICASSP submission: the test set, scoring and judge (Secs. 1–3); the statistical treatment and every robustness check (Secs. 4–7); the period ladder and the baselines (Secs. 8–9); and the state-space measurements and negative mechanistic results (Secs. 10–13).

1 The test set in full

The test set contains 180 items in five families (Table 1); scoring counts occurrences of a target unit against an expected count, not string matches against a reference transcript.

Table 1: Item families.

Family	Count	What it tests
word repetition	60	a single word repeated k times in a fixed carrier
matched control	54	the same carrier, k <i>distinct</i> filler words (no repetition)
sentence repetition	32	a whole short sentence repeated k times
tongue twisters	24	a twister sentence (with internal repetition) repeated k times
numbers	10	English number phrases with intrinsic digit-word repetition
Total	180	

The k -ladder. Word repetition and controls run $k \in \{1, 2, 3, 4, 6, 8, 12, 16, 24, 32\}$, bracketing the expected collapse point $k^* \in 4\text{--}16$; sentence repetition stops at $k = 16$, twisters at $k = 4$ (Table 4); numbers are fixed strings, no ladder (Table 3).

Word-repetition carriers. Table 2 reproduces all six carrier templates verbatim. The six targets are common, short, phonetically distinct from every filler, and homophone-free, limiting judge mis-transcription noise; the judge audit (Section 3) later located the transcription risk in the judge’s autoregressive decoder, not the word list.

Number phrases. Table 3 reproduces all ten phrases verbatim. One digit-word recurs at several magnitudes, so correct rendering needs a place-value counter, not adjacent-token runs.

Sentence repetition and tongue twisters. Sentence items (Table 4) score one content word at expected count k ; twister items expect per-copy occurrences $\times k$, the w6 and w7 targets (per-copy 0) probing phonological stress only, outside the counting metric.

Table 2: Word-repetition carriers: fixed prefix, target repeated k times, fixed suffix; the filler pool builds the matched control at the same k .

ID	Prefix	Target	Suffix	Filler pool (for the control)
t1	The dog was	very	big and it ran across the field.	quite, really, truly, fairly, rather, somewhat, extremely, notably
t2	She said	no	to the offer and then left the room.	yes, maybe, sure, fine, okay, well, right, hmm
t3	He walked	far	into the forest before turning back.	deep, fast, long, wide, high, low, near, past
t4	The light was	blue	before the storm arrived.	green, bright, pale, dim, warm, cold, sharp, soft
t5	The runner moved	quick	along the narrow path.	swift, smooth, steady, light, sharp, loose, tight, clean
t6	They were	really	tired after the long journey.	truly, quite, very, rather, deeply, clearly, plainly, surely

Table 3: Number-phrases items, with the target digit-word and its true count.

ID	Phrase	Target	Count
n01	six hundred sixty six thousand six hundred sixty six	six	6
n02	seven hundred seventy seven thousand seven hundred seventy seven	seven	6
n03	nine hundred ninety nine thousand nine hundred ninety nine	nine	6
n04	three hundred thirty three thousand three hundred thirty three	three	6
n05	six hundred sixty six million six hundred sixty six thousand six hundred sixty six	six	9
n06	eight hundred eighty eight thousand eight hundred eighty eight	eight	6
n07	five hundred fifty five	five	3
n08	four hundred forty four	four	3
n09	two hundred twenty two million two hundred twenty two thousand two hundred twenty two	two	9
n10	one hundred eleven thousand one hundred eleven	one	5

Table 4: Sentence-repetition items ($k \in \{1, 2, 3, 4, 6, 8, 12, 16\}$, expected count k) and tongue-twister items ($k \in \{1, 2, 4\}$, expected count per-copy occurrences of the target $\times k$).

ID	Sentence	Target	Per-copy
s1	The bell rang twice.	bell	—
s2	He counted the stones.	stones	—
s3	Rain fell on the roof.	rain	—
s4	The door stayed open.	door	—
w1	She sells sea shells by the sea shore.	sea	2
w2	Peter Piper picked a peck of pickled peppers.	peck	1
w3	How much wood would a woodchuck chuck.	chuck	1
w4	Red lorry yellow lorry red lorry yellow lorry.	lorry	4
w5	The sixth sick sheikh’s sixth sheep is sick.	sixth	2
w6	Six slick slim sycamore saplings.	s (alliteration only)	0
w7	Fresh French fried fly fritters.	fr (alliteration only)	0
w8	Truly rural truly rural truly rural.	rural	3

Matched controls. Same prefix, suffix, and word count, the k target copies replaced by k distinct fillers cycled from the pool, the target never appearing in its own control. Only periodicity is removed: an item-control difference cannot reflect sequence length, generation duration, or text amount, the basis of the main text’s dissociation test.

Repetition-boundary lists. Word- and sentence-repetition items list their k target copies in order, controls their k fillers; the boundary-location step (Section 10) searches one explicit, model-agnostic list, left-to-right and non-rewinding in both arms, so fitted decay rates compare directly.

The instrumented subset. Teacher-forced hidden-state and attention capture (Section 10) covers word repetition, sentence repetition, and controls at $k \geq 2$, 136 items; twisters and numbers are behavioural-only.

2 Scoring

Two flags. As built, Count A is Whisper large-v3 [1]; the judge audit (Section 3) showed its autoregressive decoder systematically undercounts correct periodic speech, so the main text uses a CTC judge scored by relative count error, and Whisper-derived numbers are flagged where they recur. A judge swap changes only the transcript source: Count B is recalibrated per judge, and every transcript-independent piece survives.

Three metrics. **Count A** counts exact whole-token target matches in the normalised transcript (lower-cased, punctuation stripped, numerals expanded to number words). **Count B** inverts a per-model, per-family, per-template linear fit of duration on k , calibrated only on a trusted low- k regime ($k \leq 3$, Count A exact, duration > 0.2 s) fixed before seeing high- k behaviour. An item is **correct** only if Count A exactly equals the expected count *and* the duration ratio against the fit lies in $[0.70, 1.45]$.

Exclusions. Audio that is empty or degenerate (duration < 0.25 s, RMS $< 10^{-3}$, spectral flatness > 0.35 , or an empty transcript) carries no count and is reported as its own rate, checked before any transcript-based label. At $|\text{Count A} - \text{Count B}| > \max(1, 0.15 \cdot \text{expected count})$ the item is a flagged disagreement, never a synthesised number.

3 The judge is part of the experiment

On ground truth known by construction, the natural instrument, Whisper large-v3, scores 0.25 of the true repetition count; the CTC recogniser that replaced it scores 1.00. Whisper’s decoder is autoregressive with a language-model prior: it discounts a further verbatim repeat and closes the sentence on repeated audio, the failure mode under study, in the instrument that must certify it. Every earlier Whisper-judged number is flagged where it recurs.

Head-to-head on synthetic ground truth. Six verified $k=1$ atoms per arm, a target word rendered exactly once and a matched control utterance with no internal repetition, each concatenated $k \in \{2, 4, 8, 16, 32\}$ times with 0.25 s silences: the true count is exact by construction, and both recognisers score identical audio. The replacement, a wav2vec2 recogniser (large, LV-60 self-trained) under connectionist temporal classification (CTC), has no autoregressive decoder and no

internal language model: per-frame character distributions, purely local, so errors cannot correlate with repetition.

Table 5: Head-to-head validation on identical synthetic ground truth. Median counted/true ratio, $k \geq 4$ pooled. The repeated-sentence arm is periodic at the sentence grain, not an aperiodic control.

Recogniser	Material	$k \geq 4$ ratio	Notes
CTC	periodic (repeated word)	1.00	22/24 exact; 2 off by 1 ($k=16, 32$)
CTC	repeated sentence (control)	1.00	23/24 exact; 1 off by 1 ($k=32$)
Whisper large-v3	periodic (repeated word)	0.19	no transcript at $k=16, 32$
Whisper large-v3	repeated sentence (control)	0.25	no transcript at $k=16, 32$

Whisper’s $k \geq 4$ ratios pool $k=4$ and $k=8$ only: every $k=16$ and $k=32$ call returned no transcript, a missing measurement, not a low ratio.

The audit enlarged from one donor system to six. Table 5 is the original audit, six donor utterances from one source model. The enlarged audit takes donor atoms from all six checkpoints, gated on presence (the CTC recogniser finds the target exactly once in the unconcatenated clip): 27 of 36 atoms pass, giving 108 trials at $k \geq 4$. There the CTC judge’s median ratio is 1.00 on every one of the six donors; Whisper scores 0.25 on the 48 trials that return a transcript and nothing on the other 60 of 108. The main text quotes this audit, so the 0.19 above and its 0.25 are two audits, not two answers.

3.1 Four independent judges on the full population

The remaining objection is that CTC’s blank/repeat collapse is exactly the confound that would produce the headline dissociation were the judge doing the counting. The same generated audio was therefore re-scored, under the paper’s own scoring and population rules, by four judges: the primary (wav2vec2 large, LV-60 self-trained), HuBERT large LS-960 (same output structure, different pretraining), wav2vec2 robust-ft, and Whisper large-v3, autoregressive with no blank-collapse rule at all.

Table 6: Four judges on the full panel population: exact rates and repeated-versus-control gap in points ($k \geq 6$, paired panel), then the gap by checkpoint and by family.

	primary	HuBERT	robust-ft	Whisper
control exact (%)	94.3	89.5	82.1	92.9
repeated exact (%)	18.2	17.9	17.5	17.1
gap (points)	+76.7	+72.3	+65.1	+75.5
Llasa-1B	78.1	78.6	65.2	69.9
Llasa-3B	77.2	74.8	65.9	76.0
Llasa-8B	81.4	80.3	69.7	71.7
Qwen3-TTS-0.6B	92.2	87.8	82.2	92.2
Qwen3-TTS-1.7B	60.0	57.8	50.0	71.1
XTTS-v2	71.1	54.4	57.8	72.2
family: Llasa	78.9	77.9	66.9	72.5
family: Qwen3-TTS	76.1	72.8	66.1	81.7
family: XTTS	71.1	54.4	57.8	72.2

Every judge’s gap is positive in 6 of 6 checkpoints and 3 of 3 families, clearing the pre-committed +50-point floor (Table 6); the weakest judge keeps 85% of the published gap. Decisive is which arm moves: across the four judges the repeated arm’s exact rate spans 1.0 point and the control arm’s 12.2 (top rows of the table); changing the scorer moves the arm the models get right and pins the arm they get wrong, the reverse of a judge-side collapse. The disagreement asymmetry agrees (+0.27 counts where the confound requires -1.0 or lower), and the one CTC-specific confound runs in our disfavour: blank-collapse merges back-to-back identical words (“VERYVERY VERRYYVERYVERY” for twelve requested copies of *very*), depressing same-word runs and never distinct-word controls, so it can inflate the gap but not manufacture it.

A cross-lingual check. The audit replicates in Spanish (same XTTS-v2 weights): on Spanish ground truth built the same way, the CTC judge scores 1.00 of the true count where Whisper large-v3 scores 0.19 and returns nothing at the two highest k . The Spanish counting arm itself failed its pre-registered rendering-quality gate (control exact rate 20.4% against a 50% floor), so the cross-lingual dissociation is reported as untested rather than replicated.

4 Checkpoint-level inference and the specification curve

The exact-rate gap is positive in six of six checkpoints and never below 60 points. The checkpoint is the sampling unit, one repeated-minus-control difference per contrast, positive meaning present; the exact-rate column feeds the main text’s headline table.

checkpoint	exact-rate (pts)	median-error (pts)	capacity
Llasa-1B [2]	+78.1	+12.5	+38.5
Llasa-3B	+77.2	+8.3	+46.5
Llasa-8B	+81.4	+16.7	+30.5
Qwen3-TTS-0.6B [3]	+92.2	+12.5	+14.0
Qwen3-TTS-1.7B	+60.0	+0.0	+8.1
XTTS-v2 [4]	+71.1	+12.5	+10.7

contrast	mean	95% CI	p	positive
exact-rate gap (pts)	76.7	[68.4, 84.4]	0.031	6/6
median-error gap (pts)	10.4	[5.6, 13.9]	0.062	5/6
capacity gap	24.7	[13.7, 36.5]	0.031	6/6

No multiplicity correction can rescue these p -values; we do not ask it to. The smallest two-sided exact p at $n = 6$ is $2/2^6 = 0.031$ and Holm’s floor $3 \times 0.031 = 0.094$, so the paper rests on effect sizes and consistency; direction generalises, magnitude does not.

4.1 The clustering, modelled instead of counted

Modelled any way that respects the clustering, the arm effect stays near 76 points and every interval excludes zero. Six checkpoints are three families, so independent replicates would double-count; gaps recomputed from the shared panel definition (Section 5) match the numbers above exactly.

Table 7: Interval constructions for the arm effect (control – repeated, percentage points). The Wald row is reported for reference and is *not* one of the six intervals the main text’s “all six” refers to.

Construction	n clusters	Estimate	95% interval	Criterion
Mixed model, crossed RE, Wald (reference only)	3 fam.	76.4	[71.9, 80.9]	z
<i>the six constructions the main text’s “all six” counts:</i>				
Mixed model, crossed RE, $t(2)$	3 fam.	76.4	[66.6, 86.3]	$t(2)$
OLS, clustered by family	3	76.2	[68.7, 83.7]	$t(2)$
OLS, clustered by checkpoint	6	76.2	[63.2, 89.2]	$t(5)$
Family-level means, t -interval	3	75.4	[65.6, 85.2]	$t(2)$
Cluster bootstrap over families (20k)	3	76.0	[71.1, 79.0]	percentile
Cluster bootstrap over checkpoints (20k)	6	76.2	[67.2, 84.8]	percentile
<i>check, not one of the six (different link function):</i>				
GEE, binomial, clustered by family	3	OR $\approx$ 96, log-odds	[3.31, 5.83]	$t(2)$

Mixed-effects and cluster-robust models. The primary specification, crossed random intercepts with a family-varying arm slope, pays its standard error out of three families, not 982 generations; Table 7 gives every construction with its clustering unit.

The main text quotes the widest of the six, the checkpoint-clustered OLS; all six exclude zero, and the GEE agrees on its different scale. **The Wald interval is excluded** as artefact: three families barely identify the family-varying arm-slope variance, so shrinkage undercuts the between-family spread.

A hierarchical Bayesian model. A binomial logit model, partially pooled: control and gap logits each carry a grand mean, family offset and nested checkpoint offset. Two independent samplers converge and agree to 0.12 points on the panel gap (Table 8).

Table 8: Posterior arm-effect gap (percentage points, control – repeated), partially pooled; all $P(\text{gap} > 0) = 1.000$ to the precision shown. Llasa [2], Qwen3-TTS [3], XTTS-v2 [4].

Level	Unit	Mean	95% credible interval
Family	Llasa	79.3	[73.4, 84.4]
Family	Qwen3-TTS	74.1	[67.7, 79.9]
Family	XTTS	73.7	[63.5, 82.2]
Checkpoint	Llasa-1B	79.5	[70.5, 86.9]
Checkpoint	Llasa-3B	77.2	[67.0, 85.5]
Checkpoint	Llasa-8B	81.0	[72.7, 88.0]
Checkpoint	Qwen3-TTS-0.6B	87.6	[80.1, 93.8]
Checkpoint	Qwen3-TTS-1.7B	60.5	[50.4, 70.0]
Checkpoint	XTTS-v2	73.7	[63.5, 82.2]

With random effects at zero the *panel* gap is 74.9 points, [41.5, 89.4], beneath the intervals above because it marginalises over the between-family variance they condition away. The main text leads with the posterior *predictive* gap for a new family (Table 9), wide enough to place the effect with high probability, not to exclude a deficit-free fourth architecture.

Prior sensitivity, both directions. The widest prior’s predictive interval includes zero, a one-in-twenty-five chance that a fourth family shows no deficit or reverses it, the cost of locating a between-family variance from three families; every prior leaves a large effect in every observed family. Table 9 refits the identical model under tighter and much wider priors.

Table 9: Prior sensitivity on the four group-level scales σ : new-family predictive gap (points).

Prior on σ	Mean	95% CrI	$P(\text{gap} > 0)$
HalfNormal(0.5), tighter, favourable	71.7	[26.6, 90.9]	1.000
HalfNormal(1.0), reported	65.3	[5.9, 93.3]	0.996
HalfNormal(3.0), much wider, unfavourable	52.3	[−1.6, 96.1]	0.956

The specification curve. 420 specifications, zero sign reversals (Table 10). Six crossed axes: exclusion rule, k threshold, cycled versus never-cycled (aperiodic) controls, seed subset, aggregation level, and outcome, exact-rate versus median-relative-error gap, scales reported separately; of 600 crossed cells, 180 duplicate populations, leaving 420, none infeasible.

Table 10: Specification curve; n is the number of specifications under that outcome definition.

Outcome	n	Range (pts)	Median	Positive	Zero / negative
Exact-rate gap	210	[34.4, 86.4]	70.5	210/210	0 / 0
Median-error gap	210	[0.0, 19.5]	8.3	197/210	13 / 0
Both outcomes combined	420	—	—	407/420	13 / 0

The exact-rate spread is $2.5\times$ and never changes sign; the median-error gap is never negative, its non-positive cells exactly zero. The main text’s 0 of 420 is exactly sign-preservation; the two ranges must not be averaged.

Table 11: Which forking choice moves the exact-rate gap (median gap at each level, percentage points), sorted by spread, descending.

Axis	Spread (pts)	Median gap by level
k threshold	22.8	$k \geq 1$: 58.0, $k \geq 6$: 74.7, $k \geq 8$: 78.3, $k \geq 12$: 80.7, $k \geq 16$: 69.4
Exclusion rule	9.7	panel: 75.8, +templates: 68.8, +cap-hits: 76.7, +degenerate: 75.7, everything: 67.1
Aggregation level	2.3	pooled: 70.6, checkpoint: 71.3, family: 69.1
Control construction	2.3	cycled: 72.0, aperiodic: 69.7
Seed subset	1.1	all seeds: 69.7, seed 0 only: 70.8

The k threshold owns the spread (Table 11): $k_{\min} = 1$ pulls in the sub-horizon range where the arms have not yet dissociated, compressing the gap mechanically. The weakest cell keeps every degenerate and cap-hit generation and stays 34 points positive.

4.2 What the brackets mean

Every bracketed pair is a 95% interval, the kind set by the estimand: Wilson score for rates, nonparametric bootstrap over templates or items for fitted quantities, the across-checkpoint range

for the $n = 6$ panel means. Item-pooled intervals ignore within-checkpoint clustering and run too narrow; the main text leans on the checkpoint-level number.

5 The shape of the deficit

Saturating versus proportional. Proportional wins in five of six checkpoints and pooled, by orders of magnitude in residual: no counting horizon within $k \leq 32$. The refuted contraction hypothesis (Section 11) would bound the largest separable repetition count by a constant N^* , so a decoder driven past that horizon should *saturate* whatever the request; median rendered count above $k = 6$ is fitted with a constant a , a proportional bk , and the interpolating $N^*(1 - e^{-k/N^*})$, free to put a horizon anywhere.

model	sat. SSE	prop. SSE	b	soft N^*	best
Llasa-1B [2]	477.5	35.0	0.855	85	proportional
Llasa-3B	464.2	79.3	0.913	119	soft horizon
Llasa-8B	353.5	2.0	0.830	71	proportional
Qwen3-TTS-0.6B [3]	499.3	1.4	0.953	338	proportional
Qwen3-TTS-1.7B	668.8	5.1	1.107	640	proportional
XTTS-v2 [4]	378.0	0.8	0.860	90	proportional
pooled	485.3	2.6	0.950	278	proportional

The soft-horizon fit, free to find a plateau, puts N^* at or past the ladder’s top in all six. That does not bear on the refuted contraction account either way: the ladder stops short of the regime a horizon would govern.

Which rows the panel may report. Four exclusions, one shared population definition.

1. **Ablation arms are not panel members:** the XTTS-v2 decoding-ablation arms (Section 7.1) are one checkpoint under altered decoding; pooling would count XTTS-v2 five times.
2. **Degenerate and empty audio has no count:** reported as its own rate, not as a large negative error.
3. **Templates whose vocabulary the judge cannot transcribe:** the CTC judge (Section 3) renders neither t2 filler, so that template scored the recogniser’s orthography; hitting both arms, the criterion cannot favour the control side.
4. **Items cut off by our own token budget:** generations that hit the harness limit have median relative error -0.44 against -0.08 , our budget truncating audio, not failed counting; removal leaves the pooled median unchanged, per-model figures not.

A fifth correction is a scoring fix, not an exclusion: the unit counter stopped at the first unit it could not find, right for repeated items, wrong for controls; it now skips, verified bit-identical on repeated items, lifting pooled control mean error $-0.150 \rightarrow -0.090$.

The extension ladder. Because the main ladder cannot tell a distant horizon from none, the same three carriers ran at $k \in \{48, 64, 96, 128\}$, with two recorded protocol departures: the token budget rises $2048 \rightarrow 8192$, and the controls draw from a 146-word pool never cycled. XTTS-v2 is not run: every $k \geq 48$ item exceeds its decoder’s ceiling, censoring counts by architecture, not horizon.

Do the exclusions fall evenly on the two arms? The one unbalanced rule removes the worst repeated generations, shrinking the reported gap. Panel rates, before any other filter:

arm	template rule	token budget	degenerate
repeated ($n = 1080$)	16.7%	10.5%	4.4%
control ($n = 972$)	16.7%	5.6%	3.7%

The template rule is exactly balanced, structurally. The budget rule hits repeated items about twice as often, as a looping model should on periodic text; excluding them shrinks the gap, keeping them would enlarge it. Degeneracy is near-balanced and small on both arms.

5.1 Fitted-form dependence of the horizon ratio

The horizon ratio’s direction is form-independent: no curve family moves any checkpoint across 1. The saturating form was adopted only after the main ladder ran proportional, so the refit uses families we did not choose: one-parameter, unit slope at $k \rightarrow 0$, asymptoting to $\hat{K}$, differing only in the bend.

A flagged repair, favourable to us: the Qwen harness counted decode steps one token short, leaving budget-truncated items inside the horizon fits; repaired and rerun, the soft-horizon ratio moves $3.5 \rightarrow 4.2$ and the three-family range $2.7\text{--}4.3 \rightarrow 3.1\text{--}5.4$, only the two Qwen checkpoints moving.

family	$\hat{c}(k)$	$\hat{K}_{\text{rep}}$	$\hat{K}_{\text{ctl}}$	ratio
soft horizon	$\hat{K}(1 - e^{-k/\hat{K}})$	24.8	104.8	4.22
hyperbolic	$\hat{K}k/(\hat{K} + k)$	33.6	181.7	5.40
tanh	$\hat{K} \tanh(k/\hat{K})$	23.7	74.6	3.14

The soft horizon is the reported fit, hyperbolic and tanh bracketing it; $\hat{K}_{\text{ctl}}$ is unchanged to the digit under the repair, matching a bug confined to the repeated arm’s truncated items. Per checkpoint:

checkpoint	soft horizon	hyperbolic	tanh
Llasa-1B [2]	5.97	8.46	4.50
Llasa-8B	4.12	5.23	3.06
Qwen3-TTS-0.6B [3]	13.13	16.33	4.90
Qwen3-TTS-1.7B	0.29	0.57	0.07

Direction is form-independent; magnitude is not. All three families agree on which checkpoints lower the horizon, the one below 1 being Section 5.3’s exception; the headline 4.2 is a point inside $3.1\text{--}5.4$, so the main text quotes the range beside it. This tests the curve choice only: every family saturates by construction.

5.2 Do our own exclusions make the gap?

Our exclusions flatter the gap by about fourteen points and do not create it; the main text gives both numbers. About a quarter of panel generations are removed by three rules, each defensible alone: three defensible rules can still sum to a selected sample.

population	n	exact rep	exact ctl	gap
panel (as reported)	1234	18.2%	94.3%	76.1
+ judge-unmeasurable template	1424	17.2%	79.8%	62.6
+ budget-truncated items	1362	16.3%	93.1%	76.7
+ degenerate audio (scored 0)	1242	17.9%	93.9%	76.0
nothing excluded at all	1620	14.8%	76.5%	61.7

Only the template rule does work, and on the *control* side: the judge emits neither t2 filler, so restored t2 rows score the recogniser’s vocabulary, judge noise on the control arm shrinking the gap. The budget rule, the one that could have flattered us, does not, truncation hitting both arms; degenerate audio scored zero is the harshest choice.

5.3 The inverting checkpoint is the panel’s weakest

Qwen3-TTS-1.7B [3], the one checkpoint that shows no repeated-side saturation within $k \leq 128$ and inverts the horizon ratio, is also last on all three measures in the table opening Section 4, and the only checkpoint last on any: the exception sits where the effect is smallest, where smallest tips into absent. Separating its two candidates, a per-repetition map that may not contract (Section 11) or a horizon past $k = 128$, needs a longer ladder than its generation budget allows. A weakest checkpoint exists in any panel, so the ranking is no evidence for the contraction account; but it rules out the worry that the horizon result rests on discarding a contradicting checkpoint.

6 Alternative explanations, answered with measurements

Five falsifiable alternative explanations, each answered with a measurement, none overturning the dissociation; the closing paragraph collects the statistical machinery, including an uncorrected count of the comparisons.

Naturalness. The control is the *less* probable text under every independent scorer: naturalness runs opposite the alternative. **Objection:** twelve copies of *very* are far more out-of-distribution than twelve distinct intensifiers (Section 1), so failure driven by improbability rather than periodicity would leave the repeated item less probable in most of the 54 pairs. **Test:** per-token cross-entropy, same 54 pairs, three scorers in the roles Table 12 names; the template-bootstrap gap intervals exclude zero for both independent scorers.

The disagreement. Scorers agree on direction and pooled size, not on shape across k : phi-2’s gap narrows and reverses at $k \geq 24$, wholly past every panel k^* of 1–8 (Table 13). Where collapse happens both agree, and only that conjunction is claimed; the harder-to-predict member, always the control, is rendered correctly.

Unit invariance. Collapse points differ by a mean 1.7 repetitions across units while token cost at collapse differs 1.34×: repetition’s pattern, not the budget’s. **Objection:** a purely acoustic

Table 12: Naturalness check, three scorers, 54 matched pairs. “Ctl. higher”: fraction of pairs in which the control has the higher (less probable) per-token cross-entropy.

Scorer	Role	Rep. mean	Ctl. mean	Gap	Ctl. higher
Llasa-1B	panel backbone	14.85	16.05	1.20	89%
phi-2	independent, primary	3.69	4.89	1.20	87%
gpt2-large	secondary cross- check	3.36	5.34	1.98	100%

attractor predicts sentence repetition, several times costlier in tokens per repetition, collapsing at much smaller k ; the contraction account predicts comparable k^* across units. **Test** (Table 13): per model, k^* , the largest k with accuracy above 0.5, for both units.

Table 13: Unit invariance, as committed. Token counts are medians at the reported k^* ; n/a where no k cleared the 0.5 threshold.

Model	k_{word}^*	tok. (word)	k_{sent}^*	tok. (sent)
Llasa-1B	0	n/a	3	351
Llasa-3B	1	209	3	457
Qwen3-TTS-0.6B	4	53	4	80
Qwen3-TTS-1.7B	8	80	4	86
XTTS-v2	4	89	4	128
XTTS-v2 (no penalty, ablation)	2	63	1	29

Llasa-8B, completed after the table was committed, returns $k_{\text{word}}^*=4$, $k_{\text{sent}}^*=3$, leaving the mean difference 1.7 unchanged; the token ratio, $1.50\times$, stays provisional. These k^* are Whisper-judged (Section 3); the conclusion needs only a consistent threshold across families.

The capacity confound. Raw, diversity-adjusted, and correct-only panel medians agree within two hundredths (Table 14): the rank gap is not monotonous audio. **Objection:** sixteen correct *verys* make audio genuinely more monotonous than sixteen distinct words; effective-rank decline could measure that surface fact. It holds only if the gap in rank growth closes once output diversity is regressed out and only correct renderings are compared. **Test:** $\mathcal{N}_{\text{eff}}$ regressed on $\log k$ with three diversity covariates of the emitted token sequence, held only for Llasa, so the other adjusted cells are inestimable, not zero; and refitted on items labelled correct (Section 2).

One flag: Llasa-1B’s correct-only 0.01 is one noisy near-zero slope driving a ratio to zero without the gap closing. Two qualifications stand: the adjusted control is well-powered for only three of six checkpoints, and the correct-only control corroborates directionally, not decisively; its Whisper-judged label (Section 3) only removes items from the repeated correct pool, leaving cells under-powered, not inflated.

Count survival in the representation. Repeated retention falls below its control in six of six checkpoints (Table 15). **Objection:** in text-only autoregressive LMs a linear probe, exactly the Lipschitz readout the contraction account constrains, decodes the true count near-perfectly at every layer: unused, not erased [5]; degradation predicts the opposite. **Test:** ridge regression predicts $\log_2 k$ from the mean hidden state over the first and last thirds of each instrumented trajectory, with leave-one-template-out cross-validation; retention, best-layer late-window over early-window

Table 14: Capacity-confound ratios (repeated/control slope of $\mathcal{N}_{\text{eff}}$ against $\log k$). n pairs are repeated-side/control-side rows entering each regression; — marks an inestimable cell (fewer than 12 usable rows, or fewer than 4 distinct k values).

Model	Raw	+diversity	correct-only
Llasa-1B	0.42	0.50	0.01 ($n=14/32$)
Llasa-3B	0.28	0.34	0.80 ($n=24/22$)
Llasa-8B	0.43	0.75	— ($n=0/0$)
Qwen3-TTS-0.6B	0.57	—	0.31 ($n=42/84$)
Qwen3-TTS-1.7B	0.78	—	0.47 ($n=64/82$)
XTTS-v2	0.60	—	0.56 ($n=22/41$)
Panel median	0.50	0.50	0.47
XTTS-v2 (no penalty, abl., not pooled)	0.89	—	— ($n=9/22$)

R^2 , cancels probe capacity and item difficulty, the matched control decoding the same quantity from non-repetitive text of equal length. Best-layer R^2 stays modest, far from the text-LM near-perfect case.

Table 15: Ridge-probe retention (late-window best-layer R^2 / early-window best-layer R^2). All entries $n=54$ items.

Model	Repeated retention	Control retention
Llasa-1B	0.97	1.09
Llasa-3B	0.87	0.99
Llasa-8B	0.96	1.09
Qwen3-TTS-0.6B	1.46	1.74
Qwen3-TTS-1.7B	0.77	1.12
XTTS-v2	1.92	2.85
Panel median	0.97	1.11
XTTS-v2 (no penalty, abl., not pooled)	1.37	1.40

A second pooling defect, found in verification. The first-reported statistics pooled the no-penalty ablation; the six panel checkpoints alone give the medians shown, repeated still lower in six of six, the ablation a consistent, separate seventh point. **Conclusion:** qualified favour for representational degradation over output policy; yet below-1 retention is degradation, not erasure, and the accounts are not exclusive.

The XTTS repetition-penalty ablation. Objection and test: XTTS-v2 ships a 5.0 repetition penalty on acoustic-token decoding (Section 4); its high k^* could be that crutch’s artifact. The ablation is the identical checkpoint at penalty 1.0, excluded from every panel-pooled statistic. **Result:** penalty off gives strictly worse counting on both arms, a wider relative dissociation, and a compressed capacity range: the paired capacity gap moves from excluding zero to crossing it, the one cell in the panel-plus-ablation set where the capacity contrast misses significance, reported, not omitted; by CTC relative count error both arms degrade too far to isolate periodicity from decoding breakdown. The selective-masking reading is withdrawn; what stands: an ablation uninformative about the dissociation, XTTS-v2 penalty-on still the conservative panel member.

Statistical treatment. **Wilson intervals** for every proportion, working at the 0% and 100% boundaries. **Bootstrap over templates, not items:** the 54 pairs are 6 templates $\times$ 9 values of k , so item resampling understates uncertainty; n accompanies every estimate. **No multiple-comparison correction; the counts, for readers to apply their own:** six per-model capacity-gain tests, up to eighteen confound cells, six probe-retention tests, nine naturalness k values under two scorers; a family-wise guarantee divides $\alpha=0.05$ by six for the primary contrast, or by the low twenties.

7 Decoding-rule variations

7.1 The repetition penalty

The strongest decoding-level rival, and the emptiest: three checkpoints never had a repetition penalty and show a 78.9-point gap. The families ship penalties an order of magnitude apart (XTTS-v2 5.0, Qwen3-TTS 1.05, Llasa none), and a penalty by construction suppresses repeated tokens; had it produced the periodicity-specific deficit, the contraction account would explain what the sampler already explains.

Architecture	Penalty	Rep. exact (%)	Ctl. exact (%)	Control
XTTS-v2	1.00	15.3	30.9	collapsed
XTTS-v2	2.00	15.6	92.2	intact
XTTS-v2	3.00	21.1	91.1	intact
XTTS-v2 (shipped)	5.00	16.7	87.8	intact
XTTS-v2	8.00	15.6	84.4	intact
Qwen3-TTS-0.6B	1.00	10.0	100.0	intact
Qwen3-TTS-0.6B (shipped)	1.05	7.8	100.0	intact
Qwen3-TTS-0.6B	1.50	10.0	80.0	intact
Qwen3-TTS-0.6B	3.00	16.7	83.3	intact

Sweeping it changes nothing that matters. The two architectures disagree on the slope’s sign, the signature of a lever not acting on the relevant quantity; penalty disabled, Qwen3-TTS-0.6B keeps a 90-point gap with no repetition penalty anywhere in the decoder. Disabling XTTS-v2’s penalty collapses its *control*, so that arm says nothing about repetition; slopes are fitted only over arms whose control survives, and Qwen3-TTS is the useful contrast.

7.2 Greedy decoding

Greedy decoding leaves the gap within three points of sampled: the dissociation lives in the conditioning, not the draw. The contraction account bounds a readout with no reference to the sampler, yet every panel generation was sampled; stochastic token choice is exactly the perturbation breaking the posited time-invariant map, and greedy removes it.

Arm	n	Exact rep.	Exact ctl.	Gap
sampled, three seeds	216	8.3%	83.3%	75.0
sampled, seed 0	72	5.6%	83.3%	77.8
greedy	72	11.1%	83.3%	72.2

Qwen3-TTS-0.6B, $k \geq 6$; greedy is deterministic, one seed being the whole arm, and the greedy-sampled pair is the one the main text’s variations table quotes. **The truncation confound, checked:** greedy on repetitive text hits a generation budget more readily, and truncated counts are censored downward, exactly what would fake a greedy deficit; the cap-hit rate is 0.0% on both arms. **Limits:** one checkpoint, one alternative decoding mode: Qwen3-TTS-0.6B’s deficit is not a sampling artifact; the panel’s could still be, and beam search remains untested.

7.3 Repetition-aware sampling

The one mitigation purpose-built for this failure leaves the deficit intact, landing above the pre-committed SURVIVES band. Sections 7.2 and 7.1 suppress a token by its frequency in history with a scalar; VALL-E 2’s Repetition Aware Sampling instead “refines the original nucleus sampling process by accounting for token repetition in the decoding history” [6].

What was implemented. A reimplementaion from the published description, no code being released: draw by nucleus sampling; if the drawn token’s repetition ratio over a $K=10$ -step window exceeds $t_r=0.1$, redraw from the untruncated distribution. Three arms on Qwen3-TTS-0.6B: no-op threshold, paper defaults, and top- $p=0$. **Three gates:** at the no-op threshold RAS reproduces stock decoding bitwise, the same decoder, not a similar one; it fires on 48.8% of repeated-item decode steps against 10.8% on controls; and the control arm holds 100.0% exact, so the gap does not survive by the mitigation breaking the controls.

Result: the deficit survives. Exact-rate gaps, $k \geq 6$, CTC judge, panel exclusions:

Arm	Rep. exact	Gap	95% CI
no-op (= stock)	7.9%	+92.1	[+86.5, +97.8]
RAS, paper defaults	5.7%	+94.3	[+89.7, +98.9]
RAS, top- $p=0$	12.4%	+87.6	[+80.9, +94.4]

The band was pre-committed from this checkpoint’s sweeps: SURVIVES at $\geq +66.7$ points, the floor across penalty and greedy arms, MITIGATES at $\leq +33.4$; RAS lands above the band’s top. **A reproduction for free:** the no-op arm reproduces the published Qwen3-TTS-0.6B rows exactly, identical per-item counts on 233 of 233 paired items. **What weakens it:** one checkpoint; two top- p settings; a description implemented, not the authors’ code; and a summation ambiguity in the published ratio, resolved in the mitigation’s favour, moving the fire rate by half but not the verdict.

8 The period ladder in full

Exact renderings rise monotonically with period in every checkpoint, and half the deficit survives at period 2, where no token is adjacent to itself. The published design never varied periodicity alone, its repeated arm confounding period 1 with verbatim token identity; this ladder varies only the period, $p \in \{1, 2, 4, 8, k\}$, all thresholds pre-committed before any audio existed.

Design. At fixed carrier, word count and requested count $k \in \{16, 24, 32\}$: $p=1$ is $A A A A \dots$ (the published repeated arm), $p=2$ is $A B A B \dots$, $p=4$ and $p=8$ cycle four and eight distinct words ($p=8$ the published cycled control), and $p=k$ is k distinct words (the published never-cycled control). $p \in \{2, 4, 8\}$ draw rotation-balanced from one shared eight-word pool and are the only vocabulary-matched arms; $p=1$ and $p=k$ inherit the published arms’ vocabularies, a reported confound. Five

carriers (the sixth is excluded panel-wide, Section 5), three seeds, three checkpoints (Llasa-1B, Qwen3-TTS-0.6B, XTTS-v2), standard exclusions: $n = 1134$ of 1395 rows.

The full ladder. Exact rate ($= k$ requested units delivered, in order, the main text’s statistic), headline ordered-scan rule:

checkpoint	$p=1$	$p=2$	$p=4$	$p=8$	$p=k$	$n(1, 2, 4, 8, k)$
Llasa-1B	3.6	54.6	83.3	92.5	75.0	28, 130, 84, 40, 44
Qwen3-TTS-0.6B	6.8	41.3	73.3	100.0	86.7	44, 179, 90, 45, 45
XTTS-v2	11.1	51.7	75.6	82.2	57.8	45, 180, 90, 45, 45
MEAN	7.2	49.2	77.4	91.6	73.2	

Every checkpoint is monotone across $p = 1, 2, 4, 8$: a graded deficit, not a step. $p=k$ sits below $p=8$ everywhere, as its vocabulary caveat predicts, outside the monotonicity claim.

Five scoring rules. $D(p) = E(8) - E(p)$ under five rules, pre-committed calls $D(2)/D(1) \geq 0.50$ PERIODICITY, ≤ 0.25 VERBATIM TOKEN IDENTITY, between GRADED; ratio pooled over checkpoints:

rule	$D(1)$	$D(2)$	$D(4)$	$D(2)/D(1)$	verdict
ordered scan (headline)	84.4	42.4	14.2	0.502	PERIODICITY
unbounded recount	82.9	60.9	20.1	0.734	PERIODICITY
strict conjunction	82.9	61.2	21.2	0.738	PERIODICITY
cap hits kept (censored bound)	81.9	43.5	13.5	0.531	PERIODICITY
by family	84.4	42.4	14.2	0.502	PERIODICITY

The magnitude of $D(2)$ is rule-dependent, the PERIODICITY verdict unanimous; “by family” reproduces the ordered scan, so four independent methods, not five. The headline ratio clears 0.50 by only 0.002, Qwen3-TTS-0.6B carrying the pooled call (per-checkpoint scan ratios 0.426, 0.630, 0.430), while both recount rules clear 0.50 on every checkpoint; the stored bootstrap interval was built on strict conjunction, not the ordered scan, so the across-rule range is the honest companion to the scan estimate.

$p=2$ by rotation. Exact rate of the decisive cell over the pool’s four rotations:

checkpoint	rot. 0	rot. 1	rot. 2	rot. 3	range
Llasa-1B	69.7	55.6	42.9	51.4	26.8
Qwen3-TTS-0.6B	24.4	48.9	38.6	53.3	28.9
XTTS-v2	57.8	55.6	53.3	40.0	17.8

A single-rotation $p=2$ arm could have read roughly 25% to 70% exact; rotation balancing is what makes the cell readable.

The disambiguation arm. If the deficit were a failure to individuate token-identical neighbours, marking each copy’s boundary should restore the count; punctuation restores 0.9% of it, and individuation is falsified. The arm holds period 1 and edits only the boundary: bare (*very very very...*, the published arm), comma (*very, very,...*) and stop (*Very. Very...*), both length-matched, and and-conjoined (*very and very...*), adding $k - 1$ words; Qwen3-TTS-0.6B and XTTS-v2 only, $n = 353$ after exclusions. Pre-committed: recovered fraction $R = (E(\text{dis}) - E(\text{bare})) / (E(8) - E(\text{bare}))$ against each checkpoint’s own $E(8)$ ceiling, $R \geq 0.50$ supported, $R \leq 0.15$ falsified. Exact rate by variant ($k \in \{16, 24, 32\}$; Qwen3 = Qwen3-TTS-0.6B, XTTS = XTTS-v2), with per-variant confound means:

variant	exact rate		recovered R		confound means		
	Qwen3	XTTS	Qwen3	XTTS	words	chars	dur. (s)
bare	6.8	11.1	—	—	31.8	170.6	11.65
comma	6.8	8.9	+0.000	−0.031	31.8	194.3	13.11
stop	9.8	13.6	+0.031	+0.036	31.6	191.6	13.42
and	15.6	11.1	+0.094	+0.000	54.8	263.0	16.23

Mean R over the length-matched cells is $R_{\text{matched}} = 0.009$, with $R_{\text{and}} = 0.047$ beside it, never averaged in: both far under the 0.15 bar, the pre-committed falsification. The and-variant nearly doubles the word count, which no front-end can silently normalise away, so its small positive R is conservative, added length independently hurting counting.

The odd steps. Periods 3 and 6, which neither a power-of-two decoder nor a pool-cycling effect predicts should interpolate, land exactly where periodicity predicts. Only $k=24$ admits such periods, so both ran there on Llasa-1B, Qwen3-TTS-0.6B, Qwen3-TTS-1.7B and XTTS-v2, 720 generations, all else as above; pre-committed before any audio, against the $k=24$ -only ladder: CONFIRMED iff $E(2) < E(3) < E(4)$ (Test A) and $E(4) < E(6) < E(8)$ (Test B) pooled, REFUTED iff either odd step reaches $E(8)$ or falls below $E(2)$.

Both tests pass: the pooled ordered-scan rates at $p = 1, 2, 3, 4, 6, 8$ are the row the main text’s period table carries, monotone across all six periods under all four counting rules and every leave-one-checkpoint-out fit. Every adjacent gap excludes zero except the saturated ladder top (Table 16), so Test B’s upper inequality is not separable from noise.

Table 16: Adjacent-step gaps in exactly-right points on the $k=24$ ladder, carrier-cluster bootstrap 95% intervals.

adjacent steps	gap (pts)	95% interval
$E(3) - E(2)$	+10.9	[+2.1, +16.8]
$E(4) - E(3)$	+12.9	[+3.6, +21.2]
$E(6) - E(4)$	+14.7	[+2.2, +28.3]
$E(8) - E(6)$	+2.7	[−2.4, +7.7]

Shuffled order at fixed composition. Destroying the periodic order while fixing each item’s filler multiset recovers most of the deficit on both Qwen checkpoints, none of it on Llasa-1B, and destroys XTTS-v2: periodic ordering drives the deficit on some architectures, not all. The test answers the rival the ladder cannot, a decoder failing on low type-token ratio rather than periodic ordering, per a pre-committed protocol. $p=2$ is uninformative by a counting argument (exactly

two orders of twelve A’s and twelve B’s avoid adjacent equal tokens, both the periodic item, so aperiodicity forces adjacent verbatim repetition) and at $p=8$ no deficit remains, so everything rests on $p=4$, where zero adjacency is constructible and was achieved exactly.

Pooled, shuffling recovers nothing: E 75.2% $\rightarrow$ 75.3% against a 17.7-point deficit, $R(4) = +0.006$, the pre-committed confirmation band excluded ($P[\bar{R} \geq 0.50] = 0.010$), written outcome INTERMEDIATE. But the pooled figure is arithmetic, not consensus: three of four per-checkpoint intervals exclude zero, in opposite directions.

checkpoint	E periodic	E shuffled	Δ	95% CI
Qwen3-TTS-1.7B	80.0	97.8	+17.8	[+5.6, +34.4]
Qwen3-TTS-0.6B	73.3	91.1	+17.8	[+6.7, +32.2]
Llasa-1B	74.1	68.9	−5.2	[−22.0, +11.3]
XTTS-v2	73.3	43.3	−30.0	[−53.3, −6.7]

Dropping XTTS-v2, $R(4) = +0.473$: a real architecture split, reported as such, not as the mean. The collapse is no artifact: every shuffled XTTS-v2 row renders clean, full-length speech, the collapse survives the order-insensitive recount, and re-rendered periodic controls are byte-identical to the published audio on all four checkpoints, matching published rates on every cell. The placebo correction, pre-committed to fire at $\Delta(8) \leq -5$ points, tripped by exactly 0.0 and is declined; corrected or not, no conclusion changes. The cost to the title: “periodicity, not length” holds here on two checkpoints, fails on a third and reverses on a fourth, while the length-matched dissociation holds on all six panel checkpoints and a held-out architecture, so the title now carries what survives everywhere (repetition, not length) and periodicity is the architecture-dependent finding.

9 The non-autoregressive baselines

One parallel decoder is immune to the deficit and one reproduces it, so autoregression is not the cause: the architectural reading of the deficit does not survive. The main text’s scope claim is about autoregressive decoders, and no non-autoregressive system had been measured; two such baselines therefore run the published repeated and control arms, beside the panel:

model	repeated exact (%)	control exact (%)	gap (pts)
Llasa-1B	13.3	91.5	+78.1
Llasa-3B	16.9	94.1	+77.2
Llasa-8B	11.8	93.2	+81.4
Qwen3-TTS-0.6B	7.8	100.0	+92.2
Qwen3-TTS-1.7B	38.9	98.9	+60.0
XTTS-v2	16.7	87.8	+71.1
<i>F5-TTS (2024)</i>	25.6	85.6	+60.0
<i>VITS (2021)</i>	65.6	58.9	−6.7

The two disagree. VITS shows no dissociation; F5-TTS shows one close to the panel’s, and neither has a generation-step recurrence. Per k band, in points:

	VITS	F5-TTS	AR panel
$k = 2-4$	+6.7	+8.9	-1.0
$k = 6-8$	-10.0	+43.3	+60.0
$k = 12-16$	-13.3	+56.7	+85.7
$k = 24-32$	+3.3	+80.0	+83.4

F5-TTS’s gap grows with k like the panel’s, its durations scale with k (median 2.6 s at $k=1$, 11.6 s at $k=32$), so it renders rather than truncates, and its controls hold above 83%: periodicity-specific there too.

VITS’s null is not a floor effect. At $k = 6-8$ VITS renders repeated 76.7% and control 66.7% exact: far off the floor, ordered the opposite way. Nor is it the stock configuration: the stock duration predictor would have to carry the count (output length is the sum of predicted phoneme durations), so it was perturbed at two settings chosen to degrade it:

arm	setting	exact rep	exact ctl	gap
stock	—	67.6%	49.1%	−18.5
duration	duration-noise scale 1.6	53.7%	20.4%	−33.3
rate	speaking rate 1.35	61.1%	31.5%	−29.6

The gap here is control minus repeated: every VITS arm is negative, repeated-side median relative error exactly zero in all three, and both perturbations lower the control side further, a judge effect silent on repetition; stock defaults were not load-bearing, and the report is the sign, not the size. The null also does the judge’s work: the same repeated words, same test set and same recogniser produce no gap, so whatever CTC blank-collapse costs, it does not create the dissociation.

What plausibly separates them. VITS predicts a duration per input token, so “how many” is carried structurally by the input; F5-TTS estimates one total duration and denoises the whole mel in parallel, so it must represent how much speech to produce, the burden an autoregressive decoder carries in its state. A hypothesis, not a result: formed after seeing the two baselines, resting on $n = 2$ systems, untested.

The intervention: supplying the duration. Handing F5-TTS the correct total duration repairs the deficit below $k = 12$ and does nothing at all above: the paper’s one intervention on the proposed mechanism. The supplied value is the median of F5-TTS’s own control renderings at the same k plus the measured reference-clip length, never a function of the repeated item’s output.

k	free exact (%)	given exact (%)	free median	given median
6	46.7	80.0	-0.167	+0.000
8	40.0	73.3	-0.125	+0.000
12	33.3	40.0	-0.083	-0.083
16	20.0	26.7	-0.062	-0.062
24	6.7	0.0	-0.167	-0.125
32	6.7	0.0	-0.094	-0.125

Below $k = 12$ exact rises from 43.3% to 76.7% and median error goes to zero, the split the main text quotes; at or above, the gain is 0.0 points: given the correct total length, the model still cannot place thirty-two repetitions inside it. The mild end is substantially a duration-estimation failure; the severe end, where the AR panel’s deficit is worst, is not. One model, $n = 15$ per cell, one seed triple, a median rather than an item-specific target: a direction, not an effect size.

10 State instrumentation

The panel is the main text’s six checkpoints; per-layer hidden states were captured for every generation the state-space sections analyse, at a per-checkpoint probe density of every layer to every fourth ($P = 9\text{--}16$ probed layers). States were captured by one teacher-forced forward pass over the realised token sequence (Llasa, XTTS-v2) or read out inside the generation loop itself (Qwen3-TTS), position t carrying, by causality, the exact state that produced token t . Instrumented generation was validated equivalent to normal generation before any measurement was interpreted: sampling is byte-for-byte identical, and captured inputs reproduce incremental generation. A seventh registry entry, XTTS-v2 with repetition penalty disabled, is an ablation of that row (Section 7.1), excluded by construction from every panel-pooled statistic.

The count probe. Ridge regression predicts $\log_2 k$ from mean-pooled hidden states under leave-one-template-out cross-validation, penalty $\alpha = 1.0$ at every layer, item family and checkpoint, never swept or tuned. No layer is fixed in advance: per checkpoint, item family and window (*early*: mean state over the trajectory’s first third; *late*: final third), the probe is fit at all P probed layers and the argmax of cross-validated R^2 reported. Selected-layer leave-one-template-out R^2 spans 0.18–0.71 across the 24 checkpoint–family–window cells ($n=54$ items each); these fits are the ones behind Section 6’s retention ratios.

11 The contraction premise, measured

This section expands the main text’s negative result. The candidate mechanism held that softmax attention over k near-identical copies flattens toward uniformity, so the decoder meets the same conditioning at every repetition boundary and the boundary-to-boundary state map is effectively autonomous; were that map a contraction (largest singular value $q < 1$), states would converge geometrically and no Lipschitz readout, the stop head included, could separate counts past an explicit horizon (the implication itself is machine-verified in the release). The hypothesis is quantitative: measure q . Here it is measured on five checkpoints, three Llasa [2] and two Qwen3-TTS [3], and rejected.

Two estimators failed first. The boundary-distance estimator fit $\log d_m = a + m \log q$ to boundary-state distances located by a text-attention centroid; its one assumption, monotone centroid motion, is false ($\approx 51\%$ of steps advance, a random walk), and it returned $\hat{q} \approx 1.00$ [0.99, 1.02] everywhere it ran. Its boundary-free replacement, windowed dispersion converted to a per-repetition rate, put $\hat{q}$ below 1 on the identical trajectories: it measures typical spread, not the worst-direction Lipschitz constant the horizon requires.

The estimator behind the reported q . Teacher-forcing the generated tokens makes the window-to-window map a deterministic, differentiable endomorphism; q is its largest singular value under a per-layer root-mean-square metric, required to survive a sweep over four lags and sub-stacks. Teacher-forcing deletes the token channel, biasing q toward the premise; independent cache perturbation biases against. Four pre-committed validation gates passed on all five checkpoints (synthetic positive control, relative error 3.7×10^{-4} ; exactly zero backward map; Jacobian-vector products against finite differences, $\leq 1.5 \times 10^{-3}$; Qwen replay cosine ≥ 0.99920).

The five-checkpoint panel. Table 17 reports the headline setting (lag τ , whole stack): 346 items (repeated plus length-matched control), 15,243 attempted measurements, 14,796 valid; the 447 excluded are accounted for below, not silently dropped. $q < 1$ occurs in 0 of 346 items on either arm; the recorded fraction below 1 is exactly 0.0, both arms, every checkpoint. Pooled, q runs 21.0–347.7, the main text’s quoted range; the most favourable cell anywhere (smallest bootstrap

lower bound over all lags, sub-stacks and arms) is Qwen3-TTS-0.6B’s 3.35, a panel-wide floor of 3.3. Contraction was never a live description of these decoders.

Table 17: Largest singular value q of the per-repetition map at the headline setting (lag τ , whole stack), per arm, with the paired repeated-versus-control test. Checkpoints as in the main text’s panel table.

checkpoint	layers	n pairs	q repeated [95% CI]	q control [95% CI]	paired p
Llasa-1B	16	29	38.1 [32.9, 51.4]	31.8 [27.3, 39.8]	0.017
Llasa-3B	28	26	118.8 [101.9, 156.5]	86.4 [77.2, 137.3]	0.269
Llasa-8B	32	29	347.7 [258.1, 393.5]	185.4 [142.1, 236.5]	0.256
Qwen3-TTS-0.6B	28	44	21.0 [19.7, 24.2]	21.9 [20.6, 23.2]	0.931
Qwen3-TTS-1.7B	28	45	24.7 [23.1, 25.7]	23.6 [21.4, 25.1]	0.177

Both arms expansive, in all sixteen cells. In all sixteen (lag $\times$ sub-stack) cells, on every checkpoint, on *both* arms, the bootstrap CI for q lies entirely at or above 1 and the fraction of items with $q < 1$ is 0.0 (Table 18). The one significant paired contrast points the *wrong* way (Llasa-1B: repetition *more* expansive than its control); the rest are null (Table 17). The premise fails on both arms: contraction was never there for periodicity to break.

Table 18: Range of q over all sixteen (lag $\times$ sub-stack) cells, per arm. Checkpoints as in the main text’s panel table.

checkpoint	repeated: range over 16 cells	control: range over 16 cells
Llasa-1B	[3.90, 78.10]	[3.50, 46.61]
Llasa-3B	[9.33, 182.80]	[7.60, 148.28]
Llasa-8B	[11.43, 499.13]	[12.28, 329.33]
Qwen3-TTS-0.6B	[3.35, 29.47]	[3.41, 28.23]
Qwen3-TTS-1.7B	[3.45, 37.83]	[3.35, 31.18]

Two facts sharpen the refutation past replication. First, q grows with scale inside Llasa: one family, same estimator, same headline cell, $38.1 \rightarrow 118.8 \rightarrow 347.7$ from 1B through 8B, so a “contraction mechanism specific to the larger checkpoints” is excluded in the rescuing direction. Second, Qwen3-TTS-1.7B, the one checkpoint with zero median count error, is exactly as expansive as the rest ($q = 24.7$ [23.1, 25.7]): whatever separates counting from miscounting here, it is not q . **Skip accounting.** Llasa-8B skipped 21.8% of its 1677 attempted measurements (236 exceeding its 700-token read window, 129 out of memory, reported distinctly); Llasa-3B skipped 3.1% (70 of 2286, all window), Llasa-1B 0.4%, and both Qwen checkpoints none. Every skipped anchor is recorded with its reason; truncation is exact on a causal model (a shorter read window changes which anchors form, not the values returned).

12 Dilution at the item level, and probing past the ladder

Within each (model, k) cell, the flattest-attention items are counted *best*: the item-level reading of Lemma 1 fails with the sign reversed. Flattening that destroys the ability to tell occurrences apart

predicts a *negative* within-cell correlation between attention flatness and count error (undercounting makes count error negative). Measured over three flatness measures fixed in advance, signed so higher is flatter and pooled by Fisher z weighted by cell size, all three run positive on repeated items with intervals excluding zero, and controls sit at zero on all three (nothing to dilute):

flatness measure	repeated	control
largest single share	+0.59 [+0.35, +0.75]	+0.01 [-0.34, +0.37]
block entropy	+0.43 [+0.16, +0.65]	+0.17 [-0.20, +0.49]
deviation from uniform	+0.55 [+0.30, +0.72]	+0.05 [-0.31, +0.39]

Only the item-level reading falls: Lemma 1 still holds in aggregate (near-uniform attention over repeated spans, entropy growing as $\log k$), but dilution does not pick which generations fail. Reverse causation stays open, a stalled decoder yielding exactly this peaked profile; only intervening on attention could separate consequence from cause.

Probing past the horizon. The contraction hypothesis forbade a Lipschitz readout from separating counts past N^* , saying nothing below (Section 11); every probe so far was fitted on the main ladder, $k \leq 32$, where it permits success. The $k = 48 \dots 128$ items were regenerated with hidden-state capture on five checkpoints, three Llasa [2] and two Qwen3-TTS [3] (XTTS-v2’s token ceiling censors every $k \geq 48$ item), with the same ridge probe and folds; twelve aperiodic control items matched on k and template give each checkpoint its own control arm. Because $k = 48 \dots 128$ spans a third of the main ladder’s $\log_2$ range, low R^2 alone proves nothing; the constant predictor is the yardstick:

checkpoint	arm	probe MAE	constant MAE	R^2
Llasa-1B	repeated $k = 2 \dots 32$	0.63	1.12	+0.62
Llasa-3B	repeated $k = 2 \dots 32$	0.86	1.12	+0.41
Llasa-8B	repeated $k = 2 \dots 32$	0.75	1.12	+0.52
Qwen3-TTS-0.6B	repeated $k = 2 \dots 32$	0.79	1.12	+0.47
Qwen3-TTS-1.7B	repeated $k = 2 \dots 32$	0.79	1.12	+0.47
Llasa-1B	repeated $k = 48 \dots 128$	0.50	0.50	−0.02
Llasa-3B	repeated $k = 48 \dots 128$	0.49	0.50	−0.02
Llasa-8B	repeated $k = 48 \dots 128$	0.43	0.50	+0.19
Qwen3-TTS-0.6B	repeated $k = 48 \dots 128$	0.37	0.50	+0.34
Qwen3-TTS-1.7B	repeated $k = 48 \dots 128$	0.45	0.50	+0.15
Llasa-1B	control $k = 48 \dots 128$	0.43	0.50	+0.26
Llasa-3B	control $k = 48 \dots 128$	0.41	0.50	+0.33
Llasa-8B	control $k = 48 \dots 128$	0.29	0.50	+0.56
Qwen3-TTS-0.6B	control $k = 48 \dots 128$	0.47	0.50	+0.10
Qwen3-TTS-1.7B	control $k = 48 \dots 128$	0.44	0.50	+0.20

All five decode the count below the horizon. Past it, the probe-MAE ratio against the constant predictor decides, ratios within ± 0.02 of 1 called indistinguishable rather than rounded either way:

checkpoint	MAE ratio vs. constant			MAE, repeated arm	
	rep.	reading	ctl.	length-only	state probe
Llasa-1B	1.01	indistinguishable	0.86	0.61	0.51
Llasa-3B	0.99	indistinguishable	0.83	0.56	0.49
Llasa-8B	0.86	recovers the count	0.59	0.45	0.43
Qwen3-TTS-0.6B	0.74	recovers the count	0.95	0.30	0.37
Qwen3-TTS-1.7B	0.90	recovers the count	0.89	0.36	0.45

Every control arm sits below 1, none in the tie band: where the count is lost, narrowness of the target range is ruled out on that checkpoint’s own data. Under the strict threshold the contraction account’s prediction holds in 1 of 5; by effect size, 2 of 5 show nothing recoverable past the horizon and 3 clearly do. We quote 1 of 5, the less favourable reading. And the last two columns void the Qwen recoveries: a predictor given only log trajectory length, no hidden states, beats the state probe on both Qwen checkpoints and loses on all three Llasa (repeated arm, $k = 48 \dots 128$, same folds). Budget-truncated trajectories are far commoner on the repeated arm (1–6 of 12 items per checkpoint, against 0–1 on the matched control; worst, 6 of Qwen3-TTS-0.6B’s 12), and truncated length is not independent of k : babble length alone carries k , no count read from state. The two Qwen recoveries are voided; the three Llasa probes beat the baseline, and their representational reading stands.

13 Causal interventions on the count

No causal conclusion stands: the intervention instrument fails its own positive control. Full-residual activation patching, overwriting the residual stream at 128 consecutive positions with a donor’s states, is too disruptive to interpret: states from an *unrelated* item move the rendered count $1.34\times$ as much as a donor that actually differs in k , and even a length-matched same- k control donor drives 44.4% of its runs into the token budget. The rank-1 replacement, transferring only the probe’s count coordinate, removes the damage and returns equivalence bounds (Table 19); but its positive control, the identical write applied to a carrier direction Qwen3-TTS-1.7B decodes perfectly (leave-one-template-out sign accuracy 1.00), moves the output by at most 1.3% of a full carrier transfer at the published magnitude, and on every readout tested the informative write is statistically indistinguishable from a pure sampler re-roll ($p \geq 0.38$). A null from an intervention transferring nothing along *any* direction says nothing about the count direction.

Table 19: Rank-1 count-coordinate patching on four checkpoints in two families [2, 3], 2,178 scored runs, thresholds pre-committed before any audio. P1 ratio: unrelated-to-informative disruption. Worst cell: largest median shift across donor kinds as a fraction of a full donor-to-receiver transfer (noise-floor bar: below 25%). Bound: 90% TOST equivalence bound on the informative cross- k cell in the same units, the smallest excludable effect.

checkpoint	P1 ratio	worst cell	bound	verdict
Qwen3-TTS-0.6B	1.38	14%	9%	readable null
Qwen3-TTS-1.7B	1.12	9%	6%	readable null
Llasa-1B	0.57	25%	20%	readable null
Llasa-8B	0.90	42%	—	cannot tell

Llasa-8B fails decisively: worst cell 42% of full transfer, a same- k donor carrying exactly the

receiver’s count disrupting as much as a cross- k one (0.246 against 0.217 median absolute shift), unchanged at doubled seeds; an arm where the right answer disrupts like the wrong one measures foreign state, not different count. The verdict, *cannot tell*, is not *no*, and the main text says so. What survives is methodological: rank-1 probe-direction patching at this magnitude is no working instrument on these decoders, and a causal null from this family is only as good as its positive and disruption controls; ours initially lacked both.

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